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Machine learning systems for optimizing audio advertisements

Abstract

Embodiments of an audio advertising optimization system are disclosed to enable optimization of audio ad play selection and audio ad content creation using machine learning techniques. In embodiments, the system uses audio processing model(s) to extract metadata about audio ads that it receives from advertisers, such as speaker voice characteristics, music characteristics, and types of call-to-action (CTA) used. As the ads are played to users by ad servers, conversion results associated with the ad plays are recorded. Machine learning model(s) are built based on the ad metadata, user metadata, listening context data, and the user conversion results to learn conversion patterns of the ads. The conversion patterns may be used to optimize the play selection of ad servers to improve conversion rates. In embodiments, the conversion patterns may be made available to ad production systems, which may use the data to optimize audio ad content.

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Background/Summary

BACKGROUND

(1) Digital advertising has been established for many decades with the advent of the Internet. The most prevalent and well-established type of digital advertising media are display ads (e.g. banner ads). As digital advertising technology has grown over recent years, creative optimization techniques have become increasingly common. For example, ad servers in modern ad delivery platforms are able to use results of previous ad impressions to adjust or tailor the display ad (e.g. the imagery, text, or discount offer presented within the ad) to target end-users. These optimizations

can substantially improve the user conversions generated by the display ad.

(2) Audio advertising is a less mature form of digital advertising. Digital audio ads may be inserted within audio content such as song playlists, podcasts, or live streaming content. Current audio ad delivery platforms do not generally provide creative optimization for audio ads. Typically, an advertiser provides several versions of an audio ad for an ad campaign, and the ad delivery platform will play the versions in an even rotation. This approach does not allow the creative content or the play selection of the audio ads to be optimized. Another distinction between audio ads and display ads is that while display ads typically use only a single type of call-to-action (CTA) to generate user conversions (a click or tap), audio ads can have many different types of CTA. For example, an audio ad may direct the user to remember a phone number, take the user directly to a product website, or add a product to a shopping cart. Current advertisers must make a guess as to the best type of CTA to engage end-user listeners, and have no easy way of changing the CTA based on conversion results.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is illustrating an embodiment of an audio ad optimization system (AAOS) that enables play selection and creative content optimization of audio ads, according to some embodiments.

(2) FIG. 2 illustrates an embodiment of the AAOS that is implemented in a multi-tenant infrastructure service provider network, according to some embodiments.

(3) FIG. 3 illustrates example types of observation data used by the AAOS to detect conversion patterns of audio ads and examples of detected conversion patterns, according to some embodiments.

(4) FIG. 4 illustrates an embodiment of a model management system that updates machine learning models used by the AAOS, according to some embodiments.

(5) FIG. 5 illustrates an example user interface that outputs conversion patterns for a group of audio ads detected by the AAOS, according to some embodiments.

(6) FIG. 6 illustrates an example user interface that outputs conversion patterns relevant to a new audio ad uploaded to the AAOS, according to some embodiments.

(7) FIG. 7 is a flowchart illustrating a process performed by the AAOS to optimize a play selection of a group of audio ads, according to some embodiments.

(8) FIG. 8 is a flowchart illustrating a process to programmatically optimize audio ad content, according to some embodiments.

(9) FIG. 9 is a block diagram illustrating an example computer system that can be used to implement one or more portions of the AAOS, according to some embodiments.

(10) While embodiments are described herein by way of example for several embodiments and illustrative drawings, those skilled in the art will recognize that embodiments are not limited to the embodiments or drawings described. It should be understood, that the drawings and detailed description thereto are not intended to limit embodiments to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope as defined by the appended claims. The headings used herein are for organizational purposes only and are not meant to be used to limit the scope of the description or the claims. As used throughout this application, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Similarly, the words “include,” “including,” and “includes” mean including, but not limited to.

(11) It will also be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first contact could be termed a

second contact, and, similarly, a second contact could be termed a first contact, without departing from the scope of the present invention. The first contact and the second contact are both contacts, but they are not the same contact.

DETAILED DESCRIPTION OF EMBODIMENTS

(12) Digital audio ads are ads played to users aurally, typically during other types of audio content such as song playlists, podcasts, or live streaming content. Current audio ad delivery platforms do not generally provide creative optimization for audio ads. Typically, an audio ad delivery platform may play audio ads from advertisers in an even rotation (e.g. randomly) or according to a set of static play selection rules. However, the platform will not optimize the play selection of the audio ads over time or provide detailed feedback to advertisers to allow the advertisers to optimize the creative content of their ads.

(13) One aspect of audio ads that would benefit from optimization is the call-to-action (CTA) used in audio ads to generate user conversions. Consumers have preferences in the types of CTAs that they interact with, for example, visiting a product website, registering for a mailing list, etc. New types of CTAs are emerging with voice assistants, where the voice assistant is able to provide additional information about the product to the consumer, answer user questions or pose questions to the user about the product through conversation, or order the product directly from the vendor. The proliferation of new types of CTAs makes it difficult for advertisers to select a single type of CTA for an audio ad. As a result, some advertisers will create multiple versions of an audio ad with different types of CTAs, such as: To learn more, go to product website To learn more, ask voice assistant to send more info about product To learn more, ask voice assistant to tell me about product To purchase, search e-commerce website for product To purchase, ask voice assistant to add product to wish list To purchase, ask voice assistant to add product to shopping cart To purchase, ask voice assistant to order product

(14) However, a large number of variations of an audio ad can frustrate consumers who will not engage with the ad if their preferred CTA is not provided in the audio ad, thereby lowering the overall conversion rate of the ad campaign. For example, some consumers may hear the CTA “ask voice assistant” but prefer to visit the product website, or other consumers may be prompted to “order” the product but are more comfortable adding the product to a wish list before ordering, etc.

(15) To solve these and other problems related to audio ad delivery systems in the state of the art, this disclosure describes embodiments of an audio ad optimization system (AAOS) that programmatically optimizes the play selection and creative content of audio ads using machine learning techniques. In some embodiments, the AAOS uses audio processing model(s) to extract metadata about audio ads that it receives from advertisers. The audio processing models may include machine learned models such as speech recognition models and natural language processing models to extract a rich set of features from the audio ad, such as speaker voice characteristics, speech content, music characteristics, the type(s) of CTA used in the audio ad, and so on.

(16) In some embodiments, when the audio ads are played to users by ad servers of the ad delivery platform, conversion results associated with individual ad impressions are recorded. The conversion results are saved with other data (e.g. the extracted ad features, user metadata about the listener, the listening context associated with the ad impression) to create input or training datasets for machine learning model(s). The machine learning model(s) consume the input or training data to detect conversion patterns of the audio ads. For example, a conversion pattern may indicate that for a particular product, audio ads featuring a female voice played to a female listener with a particular type of CTA is likely to yield a type of conversion. The detected conversion patterns may be reduced to a set of structured records and used by the ad servers to optimize play selection of the audio ads to improve conversion rates of the ads. For example, the ad servers may use the conversion patterns to select a version of an audio ad with the best CTA type for a given user and/or listening context.

(17) In some embodiments, the ad delivery platform may initially play a group of audio ads in an even rotation or randomly for an observation period without optimization (e.g. without use of any machine learned conversion patterns). During the observation period, machine learning models are built, trained, or seeded based on observation data collected during the period to learn conversion patterns. After the observation period, the ad servers will switch to an optimized play selection using conversion patterns learned by the machine learning models. In some embodiments, the system may implement a continuous feedback loop, where the machine learning model(s) or conversion patterns are repeatedly updated based on new observation data to progressively improve or maintain ad conversion performance.

(18) In embodiments, the conversion patterns may be made available to ad production systems, which may be operated by either third party advertisers or as part of the ad delivery platform itself. The conversion patterns may be used by the ad production systems to programmatically create or modify audio ads. For example, the ad production system may determine from the patterns that a type of CTA yielded particularly good conversion results for a specific type of listener, and programmatically modify an audio ad to use the CTA type to better target that type of listener. In some embodiments, the ad production system may employ a text-to-speech technology to modify or generate ad content. For example, the ad production system may automatically convert an ad from an “add to cart” CTA to an “add to shopping list” CTA, based on one or more observed conversion patterns, by generating a new script for the CTA in a text format and then producing an audio version of the CTA from the script (e.g. generating the appropriate speech and accompanying music). In some embodiments, the updating of audio ads may be performed as an automatic feedback loop to continuously improve the content of audio ads.

(19) Depending on the embodiment, the machine learning model(s) and/or conversion patterns may be generated based on different optimization scopes. For example, the system may maintain model(s) and/or generate conversion patterns based on the observation data for just a particular advertiser. As another example, the system may maintain model(s) and/or generate conversion patterns based on data of all audio ads promoting a particular type of product. In some embodiments, the optimization scope of the model(s) and/or conversion patterns may be configurable via a configuration interface of the system.

(20) In some embodiments, the conversion patterns detected by the system may be presented to users (e.g. third party advertisers) via a graphical user interface (GUI). The supplying of these insights can be important for marketers and business owners, who may want to review the insights that the system learns (e.g. the conversion patterns) and utilize this information to make future advertising or marketing decisions, including decisions regarding external systems. An individual conversion pattern may indicate a particular type of conversion result, a combination of attributes (e.g. ad features, listener features, and listener context attributes) that is correlated with the conversion result, and a pattern metric. The pattern metric may be a conversion metric or a strength score that indicates the correlation strength, likelihood, or confidence of the pattern. In some embodiments, the system may provide a search or querying capability via the GUI to allow users or client software to search for or filter conversion patterns learned by the system.

(21) As may be understood by those skilled in the art, the AAOS described herein is a specialized computer system that improves the functioning of existing computer systems that deliver digital audio ads to user engagement systems such as smartphones and voice assistant devices. The disclosed system implements a practical application to, among other things, programmatically improve play selection of audio ads and generate machine learned conversion patterns to optimize audio ad play selection and audio ad content. The disclosed features are designed to solve technical problems rooted in the computer field, and are not intended to capture any human mental and pen-and-paper processes, basic methods of organizing human activity, pure mathematical processes and formulas, and/or conventional business practices. These features and advantages of the AAOS system are described in further detail below, in connection with the figures.

(22) FIG. 1 is illustrating an embodiment of an audio ad optimization system (AAOS) that enables play selection and creative content optimization of audio ads, according to some embodiments.

(23) As shown, the figure depicts an audio ad optimization system **120**. AAOS **120** provides an ad receiving interface **122** to receive audio ad files **132** from an audio ad production system **110**. The audio ad production system **110** is a system that produces and uploads audio ads to an audio ad delivery system that implements the AAOS **120**. Depending on the embodiment, the audio ad production system **110** may be a system operated by a third party (e.g. an advertiser or a client of the ad delivery system), or a subsystem of the audio ad delivery system itself. The ad receiving interface **122** may be a programmatic interface such as an API, or an interactive interface such as a GUI.

(24) As shown, the audio ads are uploaded to the AAOS as audio ad files **132** (e.g. MP3 files). The audio ad files **132** are stored in an audio ad storage **130**, which may be a file system, a database that indexes the audio ad files **132** for fast access, or some other type of storage system. In some embodiments, the audio ads are read from the audio ad storage **130** by the ad play selection component **140** of the AAOS and played to consumer engagement systems.

(25) As shown, when the audio ad files **132** are received, the AAOS uses one or more audio processing models **134** to process **135** the files and extract audio ad metadata **136** for individual audio ads. In some embodiments, the audio processing model(s) **134** are machine learning (ML) models. The model(s) **134** may include speech recognition models that detects speech or voice characteristics of speakers in the audio ads, and/or natural language processing (NLP) models that recognizes the words and/or semantic content of the spoken script in the audio ads.

(26) The audio ad metadata may include a variety of classification features of the audio ad. Examples of extracted voice characteristics may include things such as speaker gender, speaker age, speaker accent, number of distinct speakers, speaker pace, speaker volume, voice inflection, speaker language, etc. In some embodiments, the ML model(s) **134** may identify specific content in the speech such as the use of specific words, the type of product being promoted, the type and informational content of the CTA, etc. In some embodiments, the model(s) **134** may extract additional metadata about the CTA such as the number of times a CTA is repeated in the ad, where in the ad a CTA appears (e.g. beginning, middle, or end), voice of melodic characteristics of the CTA, etc. The audio processing model(s) **134** may also extract characteristics about any background music played in the ad, including the type of music (e.g. country, rock, classical), the auditory qualities of the music (e.g. soft or loud, fast or slow, somber or upbeat, lyrical or instrumental). Additionally, the ad metadata **136** may include metadata that are not extracted using audio processing. Examples of such metadata may include the audio file name, the length or size of the audio file, the promoted product or service, the identity of the advertiser, or one or more specific artists whose content is used in the ad, etc. In some embodiments, the ad metadata **136** is encoded into a concise feature vector, which can be readily consumed by downstream machine learning models (e.g. ad selection model(s) **160**).

(27) As shown, after the audio ads are stored in the audio ad storage **130**, an ad play selection component **140** will begin selecting individual audio ads to deliver **143** to consumer engagement system **150** to create ad impressions. In some embodiments, the ad play selection component **140** is implemented by individual ad servers in a fleet of ad servers of the ad delivery system. The audio ads may be delivered via an ad delivery interface **144**, which may be a programmatic interface such as an API or service interface. The ad delivery interface **144** may receive listener context information **151b** for ad impression slots from consumer engagement systems **150** and serve up individual audio ads for the slots based on the listener context information. The audio ads may be delivered and played by consumer engagement systems during the middle of publisher content such as song lists, podcasts, live broadcasts, etc.

(28) The consumer engagement systems **150** includes a variety of computer devices that form the ad distribution network of the ad delivery system. Examples of consumer engagement systems **150**

may include music play devices, smartphones (including particular apps on smartphones), user wearable devices, personal computers, vehicles, web portals, and the like. One example of a consumer engagement system **150** is an e-commerce website that lists items for sale to users and/or plays audio ads to promote certain items. Such a website may maintain user data about individual users such as demographic information about a user, promotional preferences of the user, the purchase history of the user, etc. Another type of consumer engagement device is the voice assistant device, which is capable of interacting with the user through conversation. This conversation capability allows the voice assistant to implement different types of CTAs, for example, to answer user questions about an item or service, pose questions to the user about an item or service (e.g. asking for user opinions about a product or a related topic), manage a shopping cart of the user, or order items on behalf of the user. User responses to these CTAs may be captured by the voice assistant as conversion result data. A voice assistant device may comprise a smartphone with an installed voice assistant app. In some embodiments, the voice assistant may be integrated with or adapted to work with one or more smart speakers located in various locations near a consumer.

(29) As shown, as the delivered audio ads are played by the consumer engagement systems **150**, the consumer engagement systems may return user conversion results **151a** to the AAOS. In some embodiments, the user conversion results **151a** may be returned via the ad delivery interface **144** of the AAOS. A user conversion result **151a** may indicate a user action taken in response to a particular ad impression. Examples of user conversions may include the user actually ordering the promoted product or service, the user adding the product/service to shopping cart or a wish list, the user downloading information about the product/service, the user asking a voice assistant about the product/service, the user requesting a free trial of the product/service, the user subscribing to a user group or mailing list related to the product/service, the user commenting or sharing information about the product/service on social media, among other types of actions. In some embodiments, the conversion result **151a** may indicate a negative result, such as a failure to convert after the ad impression, or an explicit indication that the user is not interested in the promoted product or service. In some embodiments, the conversion result **151a** may indicate that the user conversion occurred after the passage of a period of time (e.g. a few hours) from the ad impression. In some embodiments, the conversion result **151a** may indicate that the conversion occurred on a consumer engagement system other than the consumer engagement system that delivered the ad impression (e.g. the audio ad played on a smartphone app but actual conversion occurred via a web portal).

(30) As shown, once an ad impression occurs, various observation data about the impression, including user conversion results **151a**, listener context information **151b**, and the ad metadata about the audio ad **151c** are gathered by the AAOS. In some embodiments, the gathered information **151a-c** are used to create observation datasets for discovering user conversion patterns **162** about the audio ads. User conversion patterns **162** may be discovered using machine learning techniques.

(31) In some embodiments, one or more ad selection models **160** may be built by using the observed data **151** as training data. The ad selection model(s) **160** may be trained using the training data to predict one or more aspects of user conversion results based on an input feature vector of different attributes in the observation data **151**. The ad selection model **160** may be built as one or more neural networks, decision trees, support vector machines, regression functions, or some other type of ML model. Once the ad selection model is sufficiently trained to learn user conversion patterns **162** about a group of audio ads, its prediction output can be used by the ad play selector **140** to make optimized ad selections decisions **161**. The optimized ad selections will be able to better target different types of users and listening contexts with different audio ads, so as to improve overall user conversion rates.

(32) In some embodiments, the ad selection model **160** may be generated using a statistical modeling technique over the gathered observation data **151**. For example, in some embodiments, a

principal component analysis (PCA) technique may be used to identify the most predictive feature combinations in the observed data (the principal components) that are correlated with certain types of conversion results. With this type of modeling, a set of the strongest observation data attributes are identified and used to generate the user conversion patterns **162**. The optimized play selection **161** may be controlled by the user conversion patterns **162**, which may be reduced to a set of structured records (e.g. JSON records) that can be programmatically understood by the ad play selection component **140**.

(33) In some embodiments where user conversion patterns **162** learned by the ad selection model **160** are not directly output by the models, a pattern output component **170** may be implemented to extract the user conversion patterns **162**. Depending on the type of model, a variety of ML model interpretation techniques may be used to extract the user conversion patterns. For example, if the ML model is a linear model, the coefficients in the learned linear function may be analyzed to determine which feature combinations are the most impactful for certain types of conversion outcomes (e.g. the strongest conversion patterns). For more sophisticated ML models, “black-box” model interpretation techniques such as individual condition expectation (ICE), permuted feature importance, and local interpretable model-agnostic explanation (LIME) may be used to probe the model with synthetic data to determine the conversion patterns. In some embodiments, the user conversion patterns may be generated based on actual ad selection output produced by the model in response to real input data.

(34) As shown, in some embodiments, the ad play selection component **140** may be configured to initially select audio ads to play based on an unoptimized selection method **141**. This unoptimized selection may be a purely random selection regardless of the selection input data, or an even rotation that aims to provide equal playing time to each audio ad in a group (e.g. a group of ad variations that promote the same product/service). However, as the ad selection model **160** and/or user conversion patterns **162** mature over time (e.g. after a sufficient amount of training), the ad play selection component **140** will switch to the optimized play selection **161** based on the detected conversion patterns. In some embodiments, the ad play selection component **140** may implement a continuous feedback loop to continuously tune the play selection of the ad play selector **140** based on new observation data **151**. For example, the ad play selector **140** may occasionally operate in an experiment mode without selection optimization **161**, to look for any new conversion patterns. In some embodiments, the ad play selector **140** may monitor the performance of its ad play selections, and trigger a model or conversion pattern update if the performance drops below a specific level. The performance may be measured using conversion metrics, which may be based on the conversion rate, or a combination of other attributes in the user conversion results **151a**.

(35) As shown, in some embodiments, the user conversion patterns **162** output by the pattern output component **170** may be provided **171** to the audio ad production system **110**, in a second feedback loop. In the second feedback loop, the audio ad production system **110** implements an ad content optimization model **180** that consumes the user conversion patterns **162** and programmatically updates the creative content of the audio ads to achieve better conversion results. Ad content optimizations may include switching to a different type of CTA in the audio ad, changing the speaker voice or background music of the audio ad, changing the spoken script of the ad to include or exclude certain phrases, etc. As with ad play selection optimization, the ad content optimization feedback loop may occur repeatedly to tune the content of the audio ads based on observation data **151** about the ad impressions.

(36) FIG. 2 illustrates an embodiment of the AAOS that is implemented in a multi-tenant infrastructure service provider network, according to some embodiments.

(37) Multi-tenant infrastructure service provider network **200** may be a private or closed system or may be set up by an entity such as a company or a public sector organization to provide one or more computing infrastructure services (such as various types of cloud-based storage) accessible via the Internet and/or other networks to clients **270** in client premises networks, in some

embodiments. Service provider network **200** may be implemented in a single location or may include numerous data centers hosting various resource pools, such as collections of physical and/or virtualized computer servers, storage devices, networking equipment and the like, needed to implement and distribute the infrastructure and services offered by the provider network **200**. In some embodiments, provider network **200** may implement various computing systems, resources, or services, such as a virtual private cloud (VPC) service, one or more compute service(s), data storage service(s) **230**, machine learning service **250**, as well as other types of services **250**. As shown in this example, the services of the provider network are used to implement components of an audio ad delivery system that delivers audio ads provided by advertiser system **274** to consumer engagement systems **150**, for example, during audio content provided by content publisher systems **272**. The components of the ad delivery system include an ad exchange service **210**, audio servers **240**, and an ad production service **220**.

(38) In various embodiments, the components illustrated in FIG. 2 may be implemented directly within computer hardware, as instructions directly or indirectly executable by computer hardware (e.g., a microprocessor or computer system), or using a combination of these techniques. For example, the components of FIG. 2 may be implemented by a system that includes a number of computing nodes (or simply, nodes), each of which may be similar to the computer system embodiment illustrated in FIG. 9 and described below. In various embodiments, the functionality of a given system or service component (e.g., a component of data storage service **230**) may be implemented by a particular node or may be distributed across several nodes. In some embodiments, a given node may implement the functionality of more than one service system component (e.g., more than one data store component).

(39) The compute service(s) implemented by service provider network **200** offer instances, containers, and/or functions according to various configurations for client operations. A virtual compute instance may, for example, comprise one or more servers with a specified computational capacity (which may be specified by indicating the type and number of CPUs, the main memory size, and so on) and a specified software stack (e.g., a particular version of an operating system, which may in turn run on top of a hypervisor). A container may provide a virtual operating system or other operating environment for executing or implementing applications. A function may be implemented as one or more operations that are performed upon request or in response to an event, which may be automatically scaled to provide the appropriate number computing resources to perform the operations in accordance with the number requests or events. A number of different types of computing devices may be used singly or in combination to implement the compute instances, containers, and/or functions of service provider network **200** in different embodiments, including general purpose or special purpose computer servers, storage devices, network devices and the like.

(40) Compute instances, containers, and/or functions may operate or implement a variety of different services, such as application server instances, general purpose or special-purpose operating systems, services that support various interpreted or compiled programming languages such as Ruby, Perl, Python, C, C++ and the like, or high-performance computing services) suitable for performing client applications, without for example requiring the client(s) **270** to access an instance. Applications (or other software operated/implemented by a compute instance and may be specified by client(s), such as custom and/or off-the-shelf software.

(41) In some embodiments, compute instances, containers, and/or functions have different types or configurations based on expected uptime ratios. The uptime ratio of a particular compute instance may be defined as the ratio of the amount of time the instance is activated, to the total amount of time for which the instance is reserved. Uptime ratios may also be referred to as utilizations in some implementations. If a client expects to use a compute instance for a relatively small fraction of the time for which the instance is reserved (e.g., 30% to 35% of a year-long reservation), the client may decide to reserve the instance as a Low Uptime Ratio instance, and pay a discounted

hourly usage fee in accordance with the associated pricing policy. If the client expects to have a steady-state workload that requires an instance to be up most of the time, the client may reserve a High Uptime Ratio instance and potentially pay an even lower hourly usage fee, although in some embodiments the hourly fee may be charged for the entire duration of the reservation, regardless of the actual number of hours of use, in accordance with pricing policy. An option for Medium Uptime Ratio instances, with a corresponding pricing policy, may be supported in some embodiments as well, where the upfront costs and the per-hour costs fall between the corresponding High Uptime Ratio and Low Uptime Ratio costs.

(42) Compute instance configurations may also include compute instances, containers, and/or functions with a general or specific purpose, such as computational workloads for compute intensive applications (e.g., high-traffic web applications, ad serving, batch processing, video encoding, distributed analytics, high-energy physics, genome analysis, and computational fluid dynamics), graphics intensive workloads (e.g., game streaming, 3D application streaming, server-side graphics workloads, rendering, financial modeling, and engineering design), memory intensive workloads (e.g., high performance databases, distributed memory caches, in-memory analytics, genome assembly and analysis), and storage optimized workloads (e.g., data warehousing and cluster file systems). Size of compute instances, containers, and/or functions, such as a particular number of virtual CPU cores, memory, cache, storage, as well as any other performance characteristic. Configurations of compute instances, containers, and/or functions may also include their location, in a particular data center, availability zone, geographic location, etc. and (in the case of reserved compute instances, containers, and/or functions) reservation term length. In this example, compute instances used by the ad exchange service **210**, audio ad servers **240**, and ad production service **220** may be provided and managed by the compute service(s) of the service provider network **200**.

(43) To implement the VPC service, the service provider network **200** provides a physical or substrate network (e.g., sheet metal boxes, cables, rack hardware) referred to as the substrate. The substrate can be considered as a network fabric containing the physical hardware that runs the services of the provider network, and can include networking devices such as routers, switches, network address translators (NATs), and so on, as well as the physical connections among the devices. The substrate may be logically isolated from the rest of the service provider network, for example it may not be possible to route from a substrate network address to an address in a production network that runs services of the service provider, or to a customer network that hosts customer resources.

(44) The VPC service may implement one or more client networks as overlay networks of virtualized computing resources (e.g., compute instances provided by the compute service(s), block store volumes, data objects such as snapshots and machine images, file storage, databases provided by the database or data storage service(s) **230**) that run on the substrate. In at least some embodiments, hypervisors or other devices or processes on the network substrate may use encapsulation protocol technology to encapsulate and route network packets (e.g., client IP packets) over the network substrate between client resource instances on different hosts within the provider network. The encapsulation protocol technology may be used on the network substrate to route encapsulated packets (also referred to as network substrate packets) between endpoints on the network substrate via overlay network paths or routes. The encapsulation protocol technology may be viewed as providing a virtual network topology overlaid on the network substrate. As such, network packets can be routed along the substrate network according to constructs in the overlay network (e.g., VPCs, security groups). A mapping service can coordinate the encapsulation and routing of these network packets. The mapping service can be a regional distributed look up service that maps the combination of overlay IP and network identifier to substrate IP so that the distributed substrate computing devices can look up where to send packets.

(45) To illustrate, each physical host can have an IP address in the substrate network. Hardware

virtualization technology can enable multiple operating systems to run concurrently on a host computer, for example as virtual machines on the host. A hypervisor, or virtual machine monitor, on a host allocates the host's hardware resources amongst various virtual machines on the host and monitors the execution of the virtual machines. Each virtual machine may be provided with one or more IP addresses in the overlay network, and the virtual machine monitor on a host may be aware of the IP addresses of the virtual machines on the host. The virtual machine monitors (and/or other devices or processes on the network substrate) may use encapsulation protocol technology to encapsulate and route network packets (e.g., client IP packets) over the network substrate between virtualized resources on different hosts within the cloud provider network. The encapsulation protocol technology may be used on the network substrate to route encapsulated packets between endpoints on the network substrate via overlay network paths or routes. The encapsulation protocol technology may be viewed as providing a virtual network topology overlaid on the network substrate. The encapsulation protocol technology may include the mapping service that maintains a mapping directory that maps IP overlay addresses (public IP addresses) to substrate IP addresses (private IP addresses), which can be accessed by various processes on the service provider network for routing packets between endpoints.

(46) In some embodiments, at least a subset of virtualization management tasks may be performed at one or more offload cards coupled to a host so as to enable more of the processing capacity of the host to be dedicated to client-requested compute instances—e.g., cards connected via PCI or PCIe to the physical CPUs and other components of the virtualization host may be used for some virtualization management components. Such an offload card of the host can include one or more CPUs that are not available to customer instances, but rather are dedicated to instance management tasks such as virtual machine management (e.g., a hypervisor), input/output virtualization to network-attached storage volumes, local migration management tasks, instance health monitoring, and the like. The offload card can function as a network interface card (NIC) of a host in some implementations, and can implement encapsulation protocols to route packets. In this example, the compute instances used by the ad exchange service **210**, audio ad servers **240**, and ad production service **220** may be connected as virtual networks provided and managed by the VPC service.

(47) As shown, the ad exchange service **210** provides a publisher interface **212** for publisher systems **272** and an advertiser interface **214** for advertiser systems **274**. These interfaces **212** and **214** may be programmatic interfaces such as API or user interfaces such as GUIs. In some embodiments, the publisher interface **212** is implemented as part of a supply side platform (SSP) that allows content publishers to register and sell content or media as inventories of advertising “space” on which audio ads can be played. In some embodiments, the advertiser interface **214** may be part of a demand side system (DSP) that allows advertisers to upload audio ads and purchase advertising space provided by the content publishers to play their audio ads. The advertiser interface **214** may implement the ad receiving interface **122** of FIG. 1. In some embodiments, the buying and selling of the advertising space may be conducted as auctions, where individual publishers ask for a certain price for a number of ad impressions and individual advertisers bid a certain price for ad impressions on certain publisher content.

(48) The data storage service **230** may be various types of data storage and processing services that perform general or specialized data storage and processing functions (e.g., analytics, big data querying, time-series data, graph data, document data, relational data, non-relational data, structured data, semi-structured data, unstructured data, or any other type of data processing operation) over data that is stored across multiple storage locations, in some embodiments. For example, the data storage service **230** may implement various types of databases (e.g., relational, NoSQL, document, or graph databases) for storing, querying, and updating data. Such services may be enterprise-class database systems that are scalable and extensible. Queries may be directed to a database in data store service **230** that is distributed across multiple physical resources, and the database system may be scaled up or down on an as needed basis, in some embodiments. The

database system may work effectively with database schemas of various types and/or organizations, in different embodiments. In some embodiments, clients/subscribers may submit queries or other requests (e.g., requests to add data) in a number of ways, e.g., interactively via an SQL interface to the database system or via APIs. In other embodiments, external applications and programs may submit queries using Open Database Connectivity (ODBC) and/or Java Database Connectivity (JDBC) driver interfaces to the database system. As shown, in this example, the data storage service **230** is used to implement the audio ad storage **130**, and also store the observation data **151** collected from ad impressions and the conversion patterns **162** generated from the observation data. (49) The machine learning service **250** may be used to perform a variety of machine learning tasks, such as preparing ML data, configuring, building, and hosting various types of ML models, performing ongoing model management such as periodic evaluation and retraining. The machine learning service **250** may offer a variety of tools for developing and monitoring ML models, such as tools to monitor the performance of a ML model and/or interpretation and explanatory tools to explain why a ML model made certain decisions. As discussed previously, interpretation and explanatory tools may be used to infer user conversion patterns **162** from black-box models that are not designed to directly output the patterns.

(50) As shown in this example, the service provider network **200** may implement ad production service **220** that allow advertisers to produce audio ads within the service provider network. The ad production service **220** may provide user interfaces that allow users to change the creative content of audio ads, for example, the script, music, CTA, and other creative aspects of the ad. In some embodiments, the ad production service **200** may be configured to change audio ad content automatically or programmatically, based on specified conditions such as ad performance or detection of particular conversion patterns. As shown, the ad production service **220** may implement audio ad production system **110** of FIG. 1, including the ad content optimization module **180** that is used to programmatically tune the content of audio ads based on observed feedback received from audio ad servers **240**. Depending on the embodiment, the ad production service **220** may be implemented as a feature of the ad exchange **210**, the ad servers **240**, or in a separate computer system external to the infrastructure service provider network **200** (e.g. an advertiser audio ad production system that integrates with the service provider network).

(51) Generally speaking, the clients **270** may encompass any type of client configurable to submit network-based requests to service provider network **200** via network **260**. For example, a given client device may include a suitable version of a web browser, or may include a plug-in module or other type of code module that may execute as an extension to or within an execution environment provided by a web browser. Alternatively, a client **270** may encompass an application such as a database application (or user interface thereof), a media application, an office application or any other application that may make use of resources in in service provider network **200** to implement various features, systems, or applications. (e.g., to store and/or access the data to implement various applications. In some embodiments, such an application may include sufficient protocol support (e.g., for a suitable version of Hypertext Transfer Protocol (HTTP)) for generating and processing network-based services requests without necessarily implementing full browser support for all types of network-based data. That is, a client **270** may be an application that interacts directly with service provider network **200**. As shown, the clients **270** in this example includes content publisher systems **272**, advertiser systems **274**, and various consumer engagement systems **150** accessible to the ad delivery system. As discussed previously, the consumer engagement systems **150** may include a variety of computer systems such as user devices **276a**, voice assistant devices **276b**, audio play devices **276c**, and web servers supporting particular websites **276d**.

(52) As shown, the clients **270** will convey network-based services requests to and receive responses from service provider network **200** via network **260**. In various embodiments, network **260** may encompass any suitable combination of networking hardware and protocols necessary to establish network-based-based communications between clients **270** and service provider network

200. For example, network **260** may generally encompass the various telecommunications networks and service providers that collectively implement the Internet. Network **260** may also include private networks such as local area networks (LANs) or wide area networks (WANs) as well as public or private wireless networks. For example, a given client **270** and service provider network **200** may be respectively provisioned within enterprises having their own internal networks. In such an embodiment, network **260** may include the hardware (e.g., modems, routers, switches, load balancers, proxy servers, etc.) and software (e.g., protocol stacks, accounting software, firewall/security software, etc.) necessary to establish a networking link between given client **270** and the Internet as well as between the Internet and service provider network **200**. It is noted that in some embodiments, clients **270** may communicate with service provider network **200** using a private network rather than the public Internet.

(53) FIG. **3** illustrates example types of observation data used by the AAOS to detect conversion patterns of audio ads and examples of detected conversion patterns, according to some embodiments.

(54) As shown, the top of FIG. **3** depicts an example observation record **310** from an observation dataset (e.g. observation data **151** of FIG. **1**). In some embodiments, each observation record **310** may correspond to a single ad impression. Each observation record reflects attributes of the particular ad impression, which may include an observation record ID **320**, ad features about the audio ad **322** (e.g. ad metadata **151c**), target user features about the user/listener who received the ad impression **324**, listening context attributes about the listening context of the ad impression **326** (e.g. listening context **151b**), and attributes of the conversion result of the ad impression **328** (e.g. user conversion results **151a**). In some embodiments, each observation record **310** may be encoded as a feature vector (e.g. a binary vector) to be consumed by ML models. The features included in the observation record **310** may vary depending on the type of ML model, and may be configurable via a configuration interface of the AAOS.

(55) As shown, the ad features **322** may include attributes such as speaker voice properties (e.g. speaker pitch, inflection, pace, and mood), particular words or phrases used by the speaker, background music properties (e.g. music type, volume, and lyrical content), the type(s) of CTA used by the audio ad, the number of times and places in the ad where a CTA was played, metadata about the audio ad file (e.g. the presence of artist content in the ad), the ID or category of the product or service that was promoted by the audio ad, and the ID or category of the advertiser that supplied the audio ad.

(56) As shown, the target user features **324** may include various attributes of the listener/user. These features may be received from the user engagement systems **150**, in a similar manner as the listening context **151b**, as discussed in connection with FIG. **1**. The user features **324** may include attributes such as user demographic data (e.g. age group, gender), one or more user segments associated with the user (e.g. high income bracket, pet owner), and one or more user-specified preferences (e.g. interested in baseball, science fiction fan). In some embodiments, the user feature data may be associated with a particular user account that is tied to multiple user engagement systems (e.g. a smartphone and a voice assistant belonging to the user). In some embodiments, the user features **324** used by the system are anonymized to not include any personally identifiable information. For example, any user level data will be processed by privacy-enabling systems so that only aggregate data are received and used by the machine learning models of the system. In this manner, the insights generated by the system will not include or be dependent on any private user information.

(57) As shown, the listening context attributes **326** may include things such as the time of the ad impression (e.g. time of day, day of week, month of year), various properties of the audio content played with the ad (e.g. category of podcast, funny content, classical music), the engagement system used to present the ad impression, the geographic location where the impression occurred, the weather at the geographic location when the impression occurred, and the frequency of the of

the ad (e.g. the number of times that the audio ad or a related ad was exposed to a user segment associated with the listener within a recent period).

(58) The conversion result **328** may simply indicate whether a conversion occurred as a result of the ad impression or not. As shown, the conversion result **328** may also include a number of attributes about a conversion, such as the time of conversion (which may be different from the time of ad impression), the time delay from the impression to conversion, the engagement system used to make the conversion (which may be different from the engagement system that provided the ad impression), the CTA type used to generate the conversion, and other types of user feedback related to the ad impression (e.g. how long the user listened to the audio ad, an indication that the user was or was not interested in the audio ad). In some embodiments, a conversion result **328** may indicate multiple user conversion actions that were caused by a single ad impression (e.g. purchase of product and comment on social media). In some embodiments, the conversion result **328** may indicate the absence of any measurable conversion, indicating that an ad exposure or impression had no effect on consumer conversion. These types of “no conversion” results may be used by the system to recognize conversion patterns about ineffective ad impressions.

(59) As discussed, from the dataset of observation records **310**, the AAOS will extract user conversion patterns, such as conversion patterns **330**. In some embodiments, the conversion patterns **330** may be stored as structured records that can be used by other software components to perform optimizations. The conversion patterns **330** may indicate a unique pattern ID **340**, a set or combination of ad play attributes **324**, a type conversion result **344** that is correlated with the combination play attributes **342**, and a pattern score **346**. The ad play attributes **342** may be any combination of ad features **322**, target user features **324**, and listening context attributes **326** recorded in the observation records **310**. The conversion result attributes **344** may be any combination of the conversion result attributes **328** recorded in the observation records **310**.

(60) It is noted that in some embodiments, a detected pattern may indicate one or more conversion result attributes **344** as independent variables and/or one or more ad play attributes **342** as dependent variables. For example, a pattern may indicate an “add to cart” CTA type of conversion strongly predicts a particular type of user engagement system. As another example, a conversion pattern may indicate that audio ads pertaining to trucks tend to be often presented with a male speaker voice. In some embodiments, the specific type(s) of the patterns detected by the AAOS is controllable by users through the configuration interface of the system.

(61) Finally, the patterns score **346** of a pattern may indicate different metrics about the pattern. In some embodiments, the score **346** may be a performance measure of the type of conversion indicated by the pattern, which may be calculated based on the underlying observation data to reflect a conversion rate, an average time to conversion, or a combination of several conversion result attributes. In some embodiments, the score **346** may reflect a strength of the correlation between the ad play attributes **342** and the conversion result attributes **344**. The strength of the correlation may be determined statistically based on the number of observation records supporting the pattern and the variance of conversion results, and may be used as an indicator of the confidence of the pattern.

(62) FIG. 4 illustrates an embodiment of a model management system that updates machine learning models used by the AAOS, according to some embodiments.

(63) As shown, FIG. 4 depicts a model management system **430** that is configured to perform periodic model updates on ad selection models **160** of the AAOS. The model updates may be used to maintain model performance over time in response to the changing nature of user conversions, or improve the performance of the models.

(64) As shown in this example, the audio ad servers **240** receive audio ads **410** from many different advertisers and delivers **420** the ads via its ad distribution network. The observation data **151** gathered by the audio ad servers **420** are periodically uploaded to the model management system **430**. In some embodiments, the audio ad servers **240** may also monitor and upload model

performance data **432** about its ad selection models. In other embodiments, the model performance data **432** may be generated by the model management system **430**.

(65) The model management system **430** operates according to model update configuration data **434** to update the ad selection models used by the audio ad servers **240**. The configuration data **434** may be received via a configuration interface of the model management system, which may be accessible to administrators of the system or ordinary users (e.g. advertisers).

(66) The model update configuration data **424** may control various aspects of how the model update management system operates. For example, as shown, the configuration data **424** may control the model scope of a particular ad selection model. The scope of the model reflects what portion of the observation data is used to build the model. For example, the bottom portion of the figure shows four ad selection models having different model scopes. Model **430** is a private model built from observation data associated with a particular advertiser A. The patterns extracted from model **430** may be visible only to advertiser A. Model **432** is built using observation data for all audio ads that promote products in product category P. Model **434** is built from observation data without taking into account a particular target user attribute (here the user segment). Model **436** is a global model that is built based on all observed data (e.g. all advertisers, product categories, and observation data attributes). The patterns extracted from model **436** may be publicly available to all advertiser users of the AAOS.

(67) In some embodiments, the configuration data **424** may define triggers and/or schedules for an ongoing model update process. Model update triggers may be defined based on the model performance data **432**. Model update schedules may define a regular schedule when model updates are performed (e.g. once a week).

(68) In some embodiments, the configuration data **424** may specify different types of model update processes for updating the models. For example, one type of model update may enable additional training of the ad selection model using new observation data. As another example, the model management system may maintain multiple versions of an ad selection model, and promote the best performing model as the production version when the current production version's performance begins to degrade. In some embodiments, the AAOS may be configured to perform a re-extraction of user conversion patterns after the model update process has stabilized, to output any new conversion patterns learned as a result of the model update.

(69) FIG. 5 illustrates an example user interface that outputs conversion patterns for a group of audio ads detected by the AAOS, according to some embodiments. The audio ad campaign report interface **510** shown in the figure may be implemented as part of the advertiser interface **212** of FIG. 2.

(70) As shown, the GUI **500** provides a campaign report for a group of audio ads that are being played in a rotation by the audio ad delivery system. The audio ads in the group are part of a single ad campaign from a single advertiser. The top portion of the GUI displays certain summary information about the campaign.

(71) As shown, section **520** displays the list of audio ads that are currently being played in the campaign. The ad version field **522** shows the name for each audio ad, which may be the name of the uploaded audio file. The ad features field **524** provides the ad features extracted for each ad, for example, the ad features **322** discussed in connection with FIG. 3. In this example, the GUI provides a clickable link to view the extracted ad features.

(72) The top conversion patterns field **526** provides a list of ad features, user features, and/or listener context attributes that are observed to yield a significant conversion result (e.g. a statistically significant jump in conversation rates compared to the rest of the ad impressions). In some embodiments, the conversion patterns field **526** may reflect a conversion pattern learned by the AAOS that is specific to one particular audio ad (e.g. extracted from observation data for the particular ad). In this example, the conversion rate field **526** indicates the conversion rate for each displayed conversion pattern.

(73) As shown, section **530** displays a list of conversion patterns that are not tied to any individual audio ads. The patterns shown in section **530** may be based on observation data learned by a global ad selection model, trained using ad impression data from a large number of different advertisers. In section **530**, the conversion pattern attributes field **534** shows a combination of ad features, user features, and listening context attributes, that predicts the conversion result field **532** (e.g. highly correlated with the conversion results). The strength field **536** shows the correlation strength of each of the detected conversion patterns.

(74) As shown, the bottom of the GUI provides a number of user controls (e.g. buttons) that allow the user to perform additional actions. The view details button may be used to view additional details about conversion patterns learned by the system (e.g. detailed statistical results of individual patterns). The download patterns button allows the user download the learned patterns as structured records (e.g. JSON records). The configure model button allows the user to make changes to parameters of the ML model(s) used to generate the conversion patterns (e.g., the model scope of the models). Finally, the modify ads button allows the user to change the set of ads used in the campaign, to add or remove ads or modify the contents of a current ad.

(75) FIG. **6** illustrates an example user interface that outputs conversion patterns relevant to a new audio ad uploaded to the AAOS, according to some embodiments. The new audio ad upload interface **610** shown in the figure may be implemented as part of the advertiser interface **212** of FIG. **2**.

(76) GUI **600** depicts a user interface generated by an embodiment of the AAOS after an audio ad file has been uploaded. The AAOS may attempt to extract audio ad metadata from the audio ad immediately after the audio file is uploaded. The top of the GUI shows some of the extracted ad metadata. The GUI also provides user control elements to add the audio ad to an ad campaign, and to correct any ad metadata that were incorrectly inferred by the AAOS (e.g. by the audio processing models **134**). In some embodiments, the user corrections may be stored by the AAOS and used to retrain or otherwise improve the performance of the audio processing models.

(77) As shown, section **620** of the GUI shows a number of conversion patterns that are relevant to the audio ad that was just uploaded. In this example, the displayed patterns all pertain to the same product category (allergy medication) as the new audio ad. The conversion patterns shown here may be similar to the conversion patterns **530** discussed in connection with FIG. **5**.

(78) As shown, the bottom of the GUI provides a number of user control elements to perform additional user actions. The filter patterns button allows the user to further filter the conversion patterns shown in section **620**, based on one or more user specified filter criteria (e.g. limit patterns shown to only female target users). The search patterns button allows the user to search for new patterns learned by the AAOS, based on one or more search criteria (e.g. show patterns with a pattern strength greater than a threshold). In some embodiments, the conversion patterns may be stored in a database system that maintains indexes for the patterns, which may be used to programmatically filter and search the patterns.

(79) Finally, in this example, the recommend publisher media button allows the AAOS to recommend one or more types of available publisher media (e.g. types of publisher content available on an ad exchange) for the audio ad. The recommendation can be based on the ad metadata and/or the relevant conversion patterns associated with the audio ad. For example, if a relevant conversion pattern suggests that the uploaded audio ad may yield good conversion results when it is played during a particular type of publisher content, publisher ad media associated with that type of publisher content will be recommended.

(80) FIG. **7** is a flowchart illustrating a process performed by the AAOS to optimize a play selection of a group of audio ads, according to some embodiments. The process may be performed by an embodiment of the AAOS **120** of FIG. **1**.

(81) The process begins at operation **710**, where a group of audio ads is received by the AAOS, to be played to target users. In some embodiments, the group of ads may be ad variations in a single

ad campaign. The ad variations may be uploaded by a single advertiser and used to promote a single product or service. In other embodiments, the group of ads may be provided by different advertisers or promote different products or services. The audio ads may be received as audio files via an ad receiving interface (e.g. ad receiving interface **122**), which may be a programmatic interface or a user interactive interface.

(82) At operation **720**, ad features or metadata of the audio ads are extracted using audio processing model(s) (e.g. audio processing models **134**). The audio processing models may include speech recognition and/or natural language processing models, and the extracted ad features may include various types of features discussed previously (e.g. audio ad metadata **136** and ad features **322**). In some embodiments, the ad metadata may be stored as individual feature vectors along with the audio ad files.

(83) At operation **730**, the audio ads are played to users via different consumer engagement systems (e.g. consumer engagement systems **150**) of the ad distribution network, and under different listening contexts (e.g. listening context **151b** classified based on listening context attributes **326**). Each play of an audio ad to a consumer creates an ad impression or exposure. The play selection of the audio ads may be determined by an ad play selection component (e.g. ad play selector **140**) of the AAOS. In some embodiments, play selection for the group of audio ads may initially be random selection, without using any conversion patterns learned about the ads.

(84) At operation **740**, user conversion results (e.g. user conversion results **151b**) of the ad impressions or exposures are received by the AASO from the consumer engagement systems. The conversion results may be received programmatically via the ad delivery interface (e.g. ad delivery interface **144**) of the AAOS, and will indicate whether a user conversion occurred as a result of an ad impression. Depending on the embodiment, the user conversion result may also indicate other types of user conversion data about the ad impression (e.g. conversion result attributes **328**). In some embodiments, the user conversion result may be returned by a different user engagement system from the user engagement system that delivered the audio ad.

(85) At operation **750**, the AAOS trains or updates machine learning models (e.g. ad selection models **160**) to learn conversion patterns (e.g. user conversion patterns **162** and **330**) from observation data associated with the ad impressions or exposures (e.g. observation data **151** formatted as observation records **310**). As discussed in connection with FIG. 3, the observation data may include attributes indicating ad features about the ad, target user features about the user, listening context attributes about the listening context, and conversion result attributes. In some embodiments, the ML model may be built using a statistical technique (e.g. PCA) that recognizes statistically significant combinations of attributes for a type of conversion result. In some embodiments, the ML model may be a type of prediction model (e.g. a neural network) that is trained to predict the conversion result based on input attributes, and the conversion patterns learned by the model may be extracted using one or more model interpretation techniques. In some embodiments, each conversion pattern may be analyzed to determine a pattern score (e.g. pattern score **346**, conversion metric **526**, or strength metric **536**). The conversion patterns learned by the ML models may be reduced to structure records such as JSON records. In some embodiments, the ML models may be managed, evaluated, updated, and interpreted by a machine learning service provided by a cloud-based infrastructure service provider network.

(86) At operation **760**, the play selection for the group of audio ads is optimized using the conversion patterns. The optimization may be performed by the audio ad servers, which may rely on the output of the ad selection model or the conversion patterns extracted or produced by the ad selection model. By using the conversion patterns, the audio ad servers will be able to better match audio ads (e.g. ads using different CTAs) to appropriate target users and/or listening contexts, in order to improve a conversion metric for the group of audio ads. As shown, in some embodiments, the process may occur repeatedly as a feedback loop, where the ad selection models, user conversion patterns, and/or ad play selection are continuously improved based on new observation

data obtained for new ad impressions.

(87) FIG. **8** is a flowchart illustrating a process to programmatically optimize audio ad content, according to some embodiments. The process may be performed by an embodiment of the AAOS **120** of FIG. **1**.

(88) At operation **810**, the AAOS uses machine learning model(s) to learn conversion patterns of audio ads based on observation data associated with ad impressions or exposures of the audio ads. Operation **810** may be performed in a fashion similar to the process discussed in connection with FIG. **7**.

(89) At operation **820**, the learned conversion patterns are generated and stored as structured data records, for example, using a component such as the pattern output component **170** of FIG. **1**. In some embodiments, each structured record may indicate a type of conversion result, a combination of ad features, user features, and/or listening context attributes that is correlated with the type of conversion result, and a pattern score of the conversion pattern (e.g. a conversion metric or correlation strength). Example structured records of conversion patterns, with example data fields, are shown in FIG. **3**. The structure records may be generated in a form that can be programmatically consumed by other software components. For example, the structure records may be generated as JSON objects. In some embodiments, the structured records may be stored in a database system, which allows the records to be filtered, searched, or queried using queries languages.

(90) At operation **830**, the structured records are output via a programmatic interface such as an API or service interface of the AAOS. The programmatic interface may be implemented as part of the advertiser interface **214** of FIG. **2**. The outputting of the conversion patterns may be performed as part of an automated process between an audio ad production system (e.g. audio ad production system **110**), and may occur periodically or be triggered as new conversion patterns are detected.

(91) At operation **840**, the conversion patterns are used (e.g. by an audio content optimization system **180**) to programmatically generate new audio ads or modify existing audio ads (e.g. audio ads being played by the AAOS). The audio ad content optimization system may be implemented as a separate system from the AAOS (e.g. a remote advertiser system), or as part of the AAOS. The audio ad content optimization system **180** may be configured to automatically generate or update audio ads to, among other things, change the speaker voice or ad background music, use different types of CTAs, change the voice or music characteristics of CTAs, etc. This content optimization may be implemented as a feedback loop to continuously tune the content of audio ads to improve conversion results.

(92) At operation **850**, the conversion patterns are used to programmatically select publisher media to deliver audio ads. In some embodiments, this type of optimization may be performed as part of audio ad play selection, where audio ads are selected for ad impressions based on the type of publisher content. In some embodiments, this optimization may be performed when a new audio ad is uploaded, as seen in FIG. **6**, where conversion patterns relevant to the new audio ad are used to recommend publisher media for purchase. Publisher media optimization may also be implemented as a feedback loop to continuously improve the quality of publisher media selection in various systems.

(93) FIG. **9** is a block diagram illustrating an example computer system that can be used to implement one or more portions of the AAOS, according to some embodiments.

(94) Computer system **1000** may include or be configured to access one or more nonvolatile computer-accessible media. In the illustrated embodiment, computer system **1000** includes one or more processors **1010** coupled to a system memory **1020** via an input/output (I/O) interface **1030**. Computer system **1000** further includes a network interface **1040** coupled to I/O interface **1030**.

(95) In various embodiments, computer system **1000** may be a uniprocessor system including one processor **1010**, or a multiprocessor system including several processors **1010** (e.g., two, four, eight, or another suitable number). Processors **1010** may be any suitable processors capable of

executing instructions. For example, in various embodiments, processors **1010** may be general-purpose or embedded processors implementing any of a variety of instruction set architectures (ISAs), such as the x86, PowerPC, SPARC, or MIPS ISAs, or any other suitable ISA. In multiprocessor systems, each of processors **1010** may commonly, but not necessarily, implement the same ISA.

(96) System memory **1020** may be configured to store instructions and data accessible by processor(s) **1010**. In various embodiments, system memory **1020** may be implemented using any suitable memory technology, such as static random access memory (SRAM), synchronous dynamic RAM (SDRAM), nonvolatile/Flash-type memory, or any other type of memory. In the illustrated embodiment, program instructions and data implementing one or more desired functions, such as those methods, techniques, and data described above, are shown stored within system memory **1020** as code **1025** and data **1035**. As shown, in some embodiments, the program instructions memory **1025** may be used to implement one or more executable components such as the ad play selection **140** and pattern output **170** components of FIG. **1**. As shown, in some embodiments, the data memory **1035** may be used to store data such as the ad metadata **136** and conversion patterns **162** of FIG. **1**.

(97) In one embodiment, I/O interface **1030** may be configured to coordinate I/O traffic between processor **1010**, system memory **1020**, and any peripheral devices in the device, including network interface **1040** or other peripheral interfaces. In some embodiments, I/O interface **1030** may perform any necessary protocol, timing or other data transformations to convert data signals from one component (e.g., system memory **1020**) into a format suitable for use by another component (e.g., processor **1010**). In some embodiments, I/O interface **1030** may include support for devices attached through various types of peripheral buses, such as a variant of the Peripheral Component Interconnect (PCI) bus standard or the Universal Serial Bus (USB) standard, for example. In some embodiments, the function of I/O interface **1030** may be split into two or more separate components, such as a north bridge and a south bridge, for example. Also, in some embodiments some or all of the functionality of I/O interface **1030**, such as an interface to system memory **1020**, may be incorporated directly into processor **1010**.

(98) Network interface **1040** may be configured to allow data to be exchanged between computer system **1000** and other devices **1060** attached to a network or networks **1050**, such as other computer systems or devices, such as routers and other computing devices, as illustrated in FIGS. **1** through **9**, for example. In various embodiments, network interface **1040** may support communication via any suitable wired or wireless general data networks, such as types of Ethernet network, for example. Additionally, network interface **1040** may support communication via telecommunications/telephony networks such as analog voice networks or digital fiber communications networks, via storage area networks such as Fibre Channel SANs, or via any other suitable type of network and/or protocol.

(99) In some embodiments, system memory **1020** may be one embodiment of a computer-accessible medium configured to store program instructions and data as described above for FIGS. **1** through **9** for implementing embodiments of methods and apparatus for traffic analysis. However, in other embodiments, program instructions and/or data may be received, sent or stored upon different types of computer-accessible media. Generally speaking, a computer-accessible medium may include non-transitory storage media or memory media such as magnetic or optical media, e.g., disk or DVD/CD coupled to computer system **1000** via I/O interface **1030**. A non-transitory computer-accessible storage medium may also include any volatile or non-volatile media such as RAM (e.g. SDRAM, DDR SDRAM, RDRAM, SRAM, etc.), ROM, etc, that may be included in some embodiments of computer system **1000** as system memory **1020** or another type of memory. Further, a computer-accessible medium may include transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as a network and/or a wireless link, such as may be implemented via network interface **1040**.

(100) Although specific embodiments have been described above, these embodiments are not intended to limit the scope of the present disclosure, even where only a single embodiment is described with respect to a particular feature. Examples of features provided in the disclosure are intended to be illustrative rather than restrictive unless stated otherwise. The scope of the present disclosure includes any feature or combination of features disclosed herein (either explicitly or implicitly), or any generalization thereof, whether or not it mitigates any or all of the problems addressed herein. Accordingly, new claims may be formulated during prosecution of this application (or an application claiming priority thereto) to any such combination of features. In particular, with reference to the appended claims, features from dependent claims may be combined with those of the independent claims and features from respective independent claims may be combined in any appropriate manner and not merely in the specific combinations enumerated in the appended claims.

(101) The methods described herein may be implemented in software, hardware, or a combination thereof, in different embodiments. In addition, the order of the blocks of the methods may be changed, and various elements may be added, reordered, combined, omitted, modified, etc. Various modifications and changes may be made as would be obvious to a person skilled in the art having the benefit of this disclosure. The various embodiments described herein are meant to be illustrative and not limiting. Many variations, modifications, additions, and improvements are possible. Accordingly, plural instances may be provided for components described herein as a single instance. Boundaries between various components, operations and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of claims that follow. Finally, structures and functionality presented as discrete components in the example configurations may be implemented as a combined structure or component. These and other variations, modifications, additions, and improvements may fall within the scope of embodiments as defined in the claims that follow.

Claims

1. A system, comprising: one or more processors with associated memory that implement an audio ad delivery system, configured to: receive, via a programmatic interface, a group of audio ad files from an advertiser for a particular product or service; process the audio ad files using one or more trained audio processing models to extract ad features about individual ones of the audio ads, including respective types of calls-to-action (CTAs) used by the audio ads identified in audio content of the audio ad files using a speech recognition model; send, via one or more audio ad servers, the audio ad files to different consumer engagement systems and under different listening contexts to play the audio ad files to create ad impressions; receive user conversion results of the ad impressions from the consumer engagement systems; train a machine learning model to learn conversion patterns for respective types of CTAs used by the audio ads in the different listening contexts; update the machine learning model based on observation data associated with the ad impressions collected by one or more audio ad servers, the observation data including the ad features, user features of the users, context attributes of the listening contexts, and the user conversion results; and automatically optimize delivery of subsequent audio ads, comprising deliver, via the one or more audio ad servers, additional plays of the audio ad files, wherein the one or more audio ad servers use the conversion patterns for respective types of CTAs in the different listening contexts to select particular versions of the audio ads for the additional plays in particular listening contexts of the listening contexts.

2. The system of claim 1, wherein the group of audio ads use different types of CTAs that call for different types of user actions with respect to a product or service, including two or more of: visiting a website associated with the product or service; subscribing to a mailing list associated

with the product or service; requesting information about the product or service; asking a voice assistant about the product or service; requesting a free trial of the product or service; adding the product or service to a wish list or shopping cart; ordering the product or service; or sharing or commenting on the product or service via a social media network.

3. The system of claim 1, wherein the audio ad delivery system is configured to: store the conversion patterns as structured records, wherein a structured record of a conversion pattern indicates a type of conversion result, a combination of ad features, user features, or context attributes that is correlated with the type of conversion result, and a pattern score associated with the conversion pattern; and output the structured records via a programmatic interface of the audio ad delivery system, wherein the structured records are used by an audio ad production system to programmatically generate new audio ads or modify one or more audio ads in the group.
4. The system of claim 1, wherein the ad features of an audio ad include one or more of: a speaker voice or a music property of a CTA in the audio ad; a number of times that the CTA is played in the audio ad; or an indication of when the CTA is played during the audio ad.
5. The system of claim 1, wherein the context attributes associated with an ad impression associated with an audio ad include one or more of: properties of publisher content during which the ad impression was created; a type of consumer engagement system used to create the ad impression; a time of the ad impression; or a frequency or number of times that the audio ad or a variation of the audio ad were exposed to one or more users prior to the ad impression.
6. The system of claim 1, wherein a user conversion result of an ad impression associated with an audio ad includes one or more of: a type of consumer engagement system that received a user conversion as a result of the ad impression; a time when the user conversion occurred; a length of time between the ad impression and the user conversion; or a type of CTA that generated the user conversion.
7. A method, comprising: performing, by one or more processors with associated memory that implement audio ad delivery system: processing audio ad files using one or more trained audio processing models to extract ad features about individual audio ads in a group of audio ads, including respective types of calls-to-action (CTAs) used by the audio ads identified in audio content of the audio ads using a speech recognition model; sending, via one or more audio ad servers, the audio ad files via different consumer engagement systems and under different listening contexts to play the audio ad files to create ad impressions; receiving user conversion results of the ad impressions from the consumer engagement systems; training a machine learning model to learn conversion patterns for respective types of CTAs used by the audio ads in the different listening contexts; updating the machine learning model based on observation data associated with the ad impressions collected by one or more audio ad servers, the observation data including the ad features, user features of the users, context attributes of the listening contexts, and the user conversion results; and automatically optimizing delivery of subsequent audio ads, comprising delivering, via the one or more audio ad servers, additional plays of the audio ad files, the one or more audio ad servers using the conversion patterns for respective types of CTAs in the different listening contexts to select particular versions of the audio ads for the additional plays in particular listening contexts of the listening contexts.
8. The method of claim 7, wherein the group of audio ads use different types of CTAs that call for different types of user actions with respect to a product or service.
9. The method of claim 7, further comprising the audio ad delivery system: storing the conversion patterns as structured records, wherein a structured record of a conversion pattern indicates a type of conversion result and a combination of ad features, user features, or context attributes that is correlated with the type of conversion result; and programmatically generating a new audio ad or modify one or more audio ads in the group based on the structured records.
10. The method of claim 7, further comprising outputting the conversion patterns via a graphical user interface (GUI).

11. The method of claim 10, wherein: the conversion patterns reflect user conversion results for audio ads of a plurality of advertisers; and the conversion patterns are provided to advertisers other than an advertiser that provided the group of audio ads.
 12. The method of claim 10, further comprising: receiving, via the GUI, a filter or search request for conversion patterns learned by the audio ad delivery system; and returning, via the GUI, one or more conversion patterns in response to the filter or search request.
 13. The method of claim 7, wherein the machine learning model is trained based on observation data associated with audio ads of a plurality of advertisers.
 14. The method of claim 7, wherein the machine learning model is periodically updated based on new observation data including new user conversion results.
 15. The method of claim 7, wherein the machine learning model is hosted by a machine learning service of an infrastructure service provider network.
 16. The method of claim 7, further comprising the audio ad delivery system: initially selecting audio ads to play from the group in without using the conversion patterns; and after the conversion patterns are detected based on observation data generated from the initial selection, using the conversion patterns to make further selections from the group.
 17. The method of claim 7, wherein: a user conversion result received by the audio ad delivery system indicates a type of CTA implemented by a voice assistant device associated with a user, wherein the voice assistant device is configured to perform: playing a first type of CTA that prompts the user to add an item to a shopping cart or an item list; playing a second type of CTA that answers one or more questions about an item or service from the user; or playing a third type of CTA that poses one or more questions to the user about an item or service, wherein user responses to the one or more questions are included in the user conversion result.
 18. One or more non-transitory computer-accessible storage media storing program instructions that when executed on one or more processors of an audio ad delivery system, cause the audio ad delivery system to: process audio ad files using one or more trained audio processing models to extract ad features about individual audio ads in a group of audio ads, including respective types of calls-to-action (CTAs) used by the audio ads identified in audio content of the audio ad files using a speech recognition model; send, via one or more audio ad servers, the audio ad files to different consumer engagement systems and under different listening contexts to play the audio ad files to create ad impressions; receive user conversion results of the ad impressions from the consumer engagement systems; train a machine learning model to learn conversion patterns for respective types of CTAs used by the audio ads in the different listening contexts; update the machine learning model based on observation data associated with the ad impressions collected by one or more audio ad servers, the observation data including the ad features, user features of the users, context attributes of the listening contexts, and the user conversion results; and automatically optimize delivery of subsequent audio ads, comprising deliver, via the one or more audio ad servers, additional plays of the audio ad files, wherein the one or more audio ad servers use the conversion patterns for respective types of CTAs in the different listening contexts to select particular versions of the audio ads for the additional plays in particular listening contexts of the listening contexts.
 19. The non-transitory computer-accessible storage media of claim 18, wherein the group of audio ads use different types of CTAs that call for different types of user actions with respect to a product or service.
 20. The non-transitory computer-accessible storage media of claim 18, wherein the program instructions when executed on the one or more processors cause the audio ad delivery system to output the conversion patterns via a graphical user interface.
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